SunCulture Kenya

Analytics Research & Portfolio Project

Business Model · Data Architecture · Power BI App Demo

Prepared by	Alex K. — Data & Analytics Engineer
Purpose	Portfolio-led job search — demonstrate end-to-end analytics engineering capability
Target role	Data & Analytics Engineer (transitioning from Senior Data Analyst)
Industry	Climate-tech / AgriFintech — SunCulture Kenya as anchor company
Date	April 2026
Deliverables	6 Power BI report pages + master App shell (HTML prototypes)

🏢 1. About SunCulture

Company overview, business model, and strategic position

1.1 What SunCulture Does

SunCulture is a Nairobi-headquartered climate-technology company founded in 2013. They design, manufacture, finance, and service solar-powered irrigation systems for smallholder farmers across East Africa. They currently operate in Kenya, Uganda, and Côte d'Ivoire, and hold a dominant position in the sub-Saharan Africa solar irrigation market.

Mission statement

Help smallholder farmers grow more food, earn more money, and build resilience to climate change — through affordable solar irrigation technology, financing, and digital services.

1.2 Core Product — RainMaker2

The flagship product is the RainMaker2 solar pump bundled with the ClimateSmart battery unit. Key specifications and SKUs:

SKU	Description & Price
RainMaker2 Basic	KES 59,999 — entry-level solar pump for small plots
RainMaker2 Pro	KES 79,999 — mid-range with expanded flow capacity
RainMaker2 Pro+	KES 104,999 — higher head pressure, larger farms
RainMaker2 Premium	KES 134,999 — largest flow rate, drip irrigation bundle
ClimateSmart battery	IoT-connected battery with embedded sensors (soil, weather)
Drip irrigation kit	Paired kit for water-efficient field distribution

1.3 Revenue Streams

SunCulture operates four revenue streams simultaneously — a critical distinction from a simple hardware company:

- PAYGrow (PAYG) installments
  - Farmers pay monthly installments over 12–36 months via M-Pesa
  - $35M+ in credit issued to 35,000+ farmers — most accessing formal credit for the first time
  - Receivables securitised via SPV (Bridgin $15M deal) — creates investor reporting obligations
- Cash hardware sales
  - 18–28% of sales — customers who pay upfront at full price
- Carbon credits (VCUs)

  - First solar irrigation carbon project in sub-Saharan Africa
  - VCUs sold to buyers including EDF Trading via Verra registry
  - Carbon revenue subsidises farmer pump cost by 25–40%
- Climate insurance (SunCulture Protect)
  - Parametric drought insurance with APA Insurance & IBISA
  - Index-based payouts triggered by rainfall shortfall data

1.4 Scale & Market Position

Metric	Value
Farmers served	60,000+ across three countries
Market share	50%+ of sub-Saharan Africa solar irrigation market
Credit portfolio	$35M+ PAYGrow loans outstanding
Employees	400+ across Kenya, Uganda, Côte d'Ivoire
Countries	Kenya (primary), Uganda, Côte d'Ivoire
Device connectivity	60,000+ internet-connected RainMaker2 units
Carbon programme	First solar-irrigation carbon project in SSA
Impact	95% of borrowers accessing formal asset finance for the first time

🗄️ 2. Enterprise Data Landscape

What data SunCulture collects across all departments

SunCulture's operations generate data across nine distinct domains. The IoT telemetry layer is the unique asset — it is the data foundation that no competitor can replicate. Below is a comprehensive inventory by department.

2.1 IoT & Device Telemetry (Streaming)

Why this matters

The ClimateSmart battery is internet-connected via Azure IoT Hub. Every device streams real-time data continuously — 60,000+ devices generating millions of events per day. This data feeds credit scoring, carbon MRV, predictive maintenance, and agronomy advisory simultaneously.

Data Category	Fields Captured
Pump operations	On/off timestamps, run-time (hrs/day), water volume pumped (L/day), flow rate, motor current draw, pump head pressure
Battery & solar	State of charge (%), solar panel voltage & wattage, charge cycles, discharge rate, battery health & degradation curve
Environmental sensors	Soil moisture, ambient temperature, rainfall (weather station), humidity, evapotranspiration estimates, drought stress flags
Device health	Online/offline status, last heartbeat, firmware version, GPS coordinates, SIM/network quality, fault error codes

2.2 Credit & PAYG Finance (Transactional)

Why this matters

SunCulture is effectively a microfinance lender. The credit data is high-stakes — errors affect underwriting decisions, investor covenant compliance, and Bridgin SPV reporting. PAR (Portfolio at Risk) must be monitored daily.

Data Category	Fields Captured
Loan origination	Application details, deposit paid, principal & term, product SKU, underwriting score, decision
Repayment tracking	M-Pesa payment date & amount, days past due (DPD), PAR 1/30/90, outstanding balance, prepayments
Collections	Call attempts & outcomes, field visit records, promise-to-pay logs, repossession events, write-offs, CRB submissions
Portfolio / SPV	Loan tape to Bridgin, collection rate by cohort, vintage analysis, write-off & recovery rates

2.3 Sales & Commercial (CRM)

Data Category	Fields Captured
Lead & pipeline	Lead source, date, region, pipeline stage, conversion rate, time-to-close, lost reason codes
Orders & fulfilment	Order date, SKU, cash vs PAYG, service center, installation date, delivery lead time
Agent performance	Agent ID & region, units sold, telesales call volume, demo conversion, commission data
Territory & market	County coverage, market penetration, farmer density by zone, targets vs actuals

2.4 Customer Success & Agronomy

Data Category	Fields Captured
Farmer profile	Farm size (acres), GPS location, crop types, water source, household demographics, prior irrigation method
Agronomy services	Soil test results, fertiliser recommendations, crop calendar advice, SMS open/act rates, farm visits, agronomist notes
Farmer outcomes	Self-reported yield change (%), income change, crop diversification events, labour hours saved, water usage before/after
Satisfaction & retention	NPS score & verbatims, CSAT by touchpoint, churn flags, referral source, support ticket volume

2.5 Carbon, Climate & MRV

Why this matters

MRV (Measurement, Reporting & Verification) requires audit-grade data lineage. Every VCU sold to EDF Trading must be backed by reproducible evidence — diesel displaced per device, CO_2e avoided, and verification period aggregates. This is a different standard of data quality from a typical BI dashboard.

Data Category	Fields Captured
Emission reduction (MRV)	Diesel displaced (litres/device), kWh generated, CO_2e avoided per farm, baseline emission factor, device active days (additionality)
Carbon registry	VCU issuance records (Verra), vintage year, buyer & transaction records, price at sale, farmer subsidy pass-through
Climate insurance	Rainfall index triggers, drought event classification, policy holder records, payout trigger dates, claim amounts
Water sustainability	Water abstraction volumes, SIWA+ limit flags, aquifer risk zone tagging, irrigation efficiency metrics (IWMI methodology)

2.6 Supply Chain, HR & Finance

Department	Key Data Fields
Supply chain	Stock on hand by SKU, serial number tracking, POs & landed cost, supplier lead times, delivery SLA actuals, spare parts stock
HR & workforce	Headcount by dept & country, hire/exit dates, attrition rate, payroll & commissions, performance scores, training completion
Finance & investor	P&L by revenue stream, multi-currency (KES/UGX/XOF), EBITDA, SPV performance to Bridgin, carbon credit timing, FX exposure, IRIS+ impact indicators

👥 3. Data Team Structure & Analytics Engineering Role

How the data function is organised and what the key roles do

3.1 Data Team Structure

SunCulture's data function sits under a Head of Data and spans three sub-functions. The team operates on Azure as the primary cloud with AWS for select workloads, and uses ClickHouse as the analytical warehouse.

Role	Responsibilities
Head of Data	Owns overall data strategy across BI, Data Engineering, and Data Science. Azure + AWS stacks.
Senior Data Scientist	Owns credit scoring ML model. Builds predictive analytics for collections, marketing, revenue. Leads AI self-service platform (LLMs/GenAI). MLOps with MLflow, Kubeflow, SageMaker.
Data & Analytics Engineer	Designs and maintains scalable data pipelines and infrastructure. Feeds the warehouse so BI dashboards and ML models have clean, reliable data. 99.99% data availability SLA.
BI Analyst	Power BI expert. Builds and optimises dashboards for all business functions. Manages RBAC. Coordinates with Data Engineers on warehouse specs. Retires unused dashboards, optimises DAX.
Data & Operations Intern	Trial performance reports, customer conversion tracking, field operations data. Reports to Director of Innovation.

3.2 Technology Stack

Stack Component	Technology
Cloud (primary)	Microsoft Azure — IoT Hub, Event Hubs, Azure AI, Azure Functions
Cloud (secondary)	AWS — Lambda, S3, SageMaker for MLOps
Analytical warehouse	ClickHouse — OLAP database for fast analytical queries
ML platform	MLflow, Kubeflow — model training, versioning, and serving
BI tool	Power BI — dashboards for all business functions + Amazon QuickSight
Languages	Python (data engineering, ML), Node.js / TypeScript (applications)
Infrastructure	Docker, Kubernetes, GitHub Actions (CI/CD)
IoT firmware	Azure AI-powered edge firmware on ClimateSmart battery
Prototyping	Streamlit, Gradio — rapid ML application prototyping

3.3 Data & Analytics Engineer — Specific Responsibilities

Based on actual SunCulture job postings, this role owns four core domains:

- IoT pipeline engineering
  - Ingesting real-time telemetry from 60,000+ connected devices
  - Building streaming pipelines: Azure Event Hubs → ClickHouse
  - Handling intermittent connectivity from rural edge devices
  - Data quality assurance for sensor readings (noise, dropouts, firmware errors)
- Credit & PAYG data infrastructure
  - Loan repayment events, PAR calculations, arrears tracking
  - Bridgin SPV integration — real-time loan tape API to warehouse
  - High-stakes data — errors affect credit decisions and investor reporting
- Carbon MRV pipelines
  - Aggregating device usage data into verifiable emission reduction calculations
  - Audit-grade data lineage and traceability for Verra registry submissions
  - Third-party verifier reproducibility requirement
- BI & ML data enablement
  - Defining warehouse schemas in ClickHouse, building dbt transformation layers
  - Coordinating with BI Analyst on data formats, structures, and field definitions
  - Enforcing 99.99% data availability SLA for dashboard consumption
  - Serving clean feature data to the Credit Scoring ML model

🏗️ 4. Data Architecture — End-to-End Design

The proposed data engineering architecture for the portfolio project

4.1 Architecture Overview

The architecture follows a modern data stack pattern: streaming ingestion from IoT devices and transactional systems → cloud storage → transformation layer → analytical warehouse → BI serving layer + ML feature store.

Design principle

A single "farmer" entity spans device telemetry, loan records, agronomy visits, and carbon MRV data. The data model must cleanly join these domains through a shared farmer dimension key — this is the core modelling challenge that demonstrates analytics engineering skill.

4.2 Ingestion Layer (Source Systems)

Source System	Ingestion Pattern
Azure IoT Hub	Device telemetry from 60K+ RainMaker2 units → Azure Event Hubs → ClickHouse time-series tables
M-Pesa payments API	Real-time repayment events → loan management system → warehouse (critical: near-zero latency for PAR)
CRM / field ops	Sales pipeline, agent activity, installation records → batch ETL daily
Verra registry (carbon)	VCU issuance confirmations → MRV pipeline → carbon fact table
Bridgin SPV platform	Loan tape API sync — two-way: SunCulture pushes portfolio data, Bridgin returns investor metrics
ERP / accounting	Financial transactions, inventory, HR → batch EL daily into warehouse
IBISA insurance platform	Parametric payout triggers, policy data → insurance fact table

4.3 Transformation Layer

Layer	What Happens Here
Raw / staging	Land data as-is from source systems — no transformations. Partition by source and date.
Cleaned / conformed	Deduplication, null handling, type casting, currency normalisation (KES/UGX/XOF → USD). Device fault code standardisation.
Business logic	PAR 30/90 calculations, carbon emission reduction formulas, sales conversion rates, DPD bucket assignments.
Data models	Star schema: fact_repayments, fact_device_events, fact_sales_orders, fact_carbon_credits + dim_farmer, dim_product, dim_date (4-4-5 calendar), dim_geography.

Tool	dbt (data build tool) — models, tests, documentation, lineage. Scheduled via Airflow or Azure Data Factory.

4.4 Serving Layer

Layer	Use Case
Power BI (Import mode)	Scheduled refresh from ClickHouse for slower-changing reports (daily). Optimised DAX, pre-aggregated tables.
Power BI (DirectQuery)	Live queries for real-time operational views (IoT device status, intraday collections).
ML feature store	Pre-computed farmer features (device usage patterns, payment history) served to credit scoring model in MLflow.
ClickHouse OLAP	Direct SQL access for data science and ad-hoc analytics. Columnar storage for fast aggregations.
RBAC / RLS	Row-level security in Power BI — regional managers see only their county data; Bridgin investors see only SPV portfolio slice.

📊 5. Power BI App — Portfolio Demo

The 6-report app built as the portfolio project centrepiece

The portfolio deliverable is a fully interactive HTML prototype simulating a Power BI App — a single workspace with a sidebar navigation, dark topbar, and six embedded report pages covering each department. All data is dynamically generated on load and refreshable.

File structure

index.html (master app shell) + dashboards/01_iot_device_ops.html, 02_credit_payg.html, 03_sales_commercial.html, 04_customer_agronomy.html, 05_carbon_climate.html, 06_executive_summary.html

5.1 Report Pages — Detailed Breakdown

Page 1 — IoT & Device Operations

Attribute	Detail
Audience	Engineering team, field operations, product management
Primary data source	Azure IoT Hub telemetry — ClickHouse time-series tables
KPI cards (8)	Active devices, online rate, avg daily runtime (hrs), avg volume pumped (L), fault rate (%), battery health, solar efficiency, firmware currency
Charts	Runtime trend (line), volume pumped trend (bar), fault rate trend (line + target), device health radar (6 dimensions)
Table / visual	Fault type breakdown (horizontal bars by count), county device grid (tile map with online rate)
Key analytics question	Which devices and counties are underperforming? Are faults trending up or down?

Page 2 — Credit & PAYG Finance

Attribute	Detail
Audience	Credit team, finance, Bridgin SPV investors
Primary data source	Loan management system, M-Pesa API, Bridgin platform
KPI cards (8)	Portfolio value (KES M), active loans, avg loan size, collection rate, PAR 30, PAR 90, first-time credit %, recovery rate
PAR gauges (4)	PAR 30, PAR 90, monthly collection rate, write-off rate — each with colour-coded status vs target
Charts	PAR 30 trend vs 10% target (line), monthly disbursements (bar), collection rate trend vs 90% target (line), product-level PAR vs collection dual-axis (bar)

Table	DPD bucket breakdown (current/1-30/31-60/61-90/90+ days), county loan performance with health badge (Good/Watch/At risk), product performance summary
Key analytics question	Which counties and SKUs are driving PAR? Is the portfolio improving or deteriorating?

Page 3 — Sales & Commercial

Attribute	Detail
Audience	Sales team, regional managers, commercial leadership
Primary data source	CRM system, telesales platform
KPI cards (8)	Units sold YTD, revenue YTD, avg selling price, lead-to-close rate, active agents, avg units/agent, field conversion rate, referral conversion rate
Charts	Monthly units vs target (bar + line), revenue trend (area line), SKU revenue mix (doughnut), lead source breakdown (doughnut)
Pipeline funnel	5-stage funnel: Leads → Qualified → Demo → Proposal → Close, with stage conversion rates
Channel cards	4 channel performance tiles: field agent, telesales, service center walk-in, referral — with units and conversion rate
County bars	Target attainment bars per county with agent count — colour-coded green/amber/red vs target
Key analytics question	Which channels and counties are over/under-performing? Where should we allocate more agents?

Page 4 — Customer Success & Agronomy

Attribute	Detail
Audience	Customer success team, agronomy team, impact investors
Primary data source	Farmer survey, agronomy visit records, IoT activity data, SMS platform
KPI cards (8)	Total farmers, active farmers, NPS, CSAT, SMS open rate, farm visits/month, soil tests, climate resilience index
Impact outcome cards (5)	Yield increase %, water usage reduction %, income uplift %, labour hours saved/week, climate resilience score
NPS display	NPS trend line + current NPS gauge with -100 to +100 scale and colour-coded marker
Charts	Farmer base growth trend (area line), churn reasons (doughnut), agronomy activity 3-panel (visits / soil tests / SMS open rate)
Other visuals	CSAT star rating breakdown (horizontal bars), crop diversification (horizontal bars by farmer count)
Key analytics question	Are farmers seeing real outcomes? Is NPS improving? What drives churn?

Page 5 — Carbon, Climate & Impact

Attribute	Detail
Audience	Carbon team, ESG investors, EDF Trading, Verra auditors
Primary data source	IoT device telemetry (MRV source), Verra registry, IBISA insurance platform, IWMI SIWA+ data
KPI cards (8)	VCUs issued YTD, VCUs sold, avg VCU price ($USD), total CO_2 avoided (tCO_2e), diesel displaced (litres), farmer subsidy per unit (KES), insurance policies, payout triggers activated
Big stat banners (4)	Total CO_2 avoided, carbon revenue generated (KES M), diesel fuel avoided (litres), water sustainability score (%)
Charts	CO_2 avoided per month (bar), VCU issuance trend (line), carbon subsidy per unit trend (line), water sustainability vs safe limit (dual-line)
Other visuals	Emission reduction by source (horizontal bars: diesel / grid / transport), carbon revenue waterfall (gross → fees → net farmer subsidy), drought event log table
Key analytics question	Is MRV data audit-ready? What is the per-unit carbon economics? Are water abstraction levels sustainable?

Page 6 — Executive Summary

Attribute	Detail
Audience	CEO, CFO, Board of Directors, impact investors
Primary data source	All domains — aggregated cross-functional view
Hero KPI cards (6)	Revenue vs KES 2.4B target (progress bar), farmers vs 70K target, market share %, NPS vs 70 target, PAR 30 vs 10% target, CO_2 avoided vs 72K target
Strategic KPI scorecard	Table of 8 strategic KPIs with current value, target, and traffic-light badge (On track / Watch / Behind)
Charts	Revenue trend (area line), farmer growth vs target (dual line), PAR 30 trend vs target (line)
Revenue mix	Doughnut chart: PAYG installments / Cash hardware / Carbon credits / Insurance & services
Department snapshots	6 department tiles (IoT, Credit, Sales, Customer, Carbon, Supply Chain) each showing 3 top metrics
Key analytics question	Are we on track to hit the year's strategic targets? Which function needs leadership attention?

🎯 6. Portfolio Narrative — How to Present This

The story to tell when showcasing this project in job applications

6.1 The Problem Statement

SunCulture operates across five fundamentally different data domains — real-time IoT telemetry, microfinance credit data, commercial CRM, agronomy field data, and carbon MRV — none of which speak to each other natively. A farmer who uses their pump every day is a credit risk signal, a carbon credit generator, and an agronomy advisory target simultaneously. Without a unified data warehouse, these insights stay siloed.

6.2 The Solution You Built

- Designed a star schema data model with a shared farmer dimension key that connects device, loan, agronomy, and carbon records
- Mapped ingestion patterns for each source system (IoT Hub streaming, M-Pesa API events, CRM batch ETL, Verra registry sync)
- Defined dbt transformation layers for business logic (PAR calculations, CO_2e formulas, DPD bucket logic, sales conversion)
- Built a 6-page Power BI App (prototyped in HTML/JavaScript) with department-specific reports and an executive summary
- Implemented RBAC logic — regional managers see county data, Bridgin investors see SPV slice, carbon auditors see MRV lineage

6.3 Skills Demonstrated

Skill	How Demonstrated
Data modelling	Star schema design, shared dimensions, fact table grain decisions, slowly changing dimensions for farmer profiles
Pipeline engineering	Streaming ingestion pattern (IoT Hub → Event Hubs → ClickHouse), batch ETL design, API integration patterns
Data quality	Handling intermittent IoT connectivity, sensor noise, currency normalisation, audit-grade carbon MRV lineage
BI development	Power BI report design across 6 audience types, DAX measure design, RBAC/RLS implementation
Domain knowledge	PAYG credit, PAR calculations, carbon MRV methodology (Verra), IoT telemetry patterns, FMCG-style commercial analytics
Communication	Translating complex multi-domain data into audience-appropriate reports (field ops vs CFO vs carbon auditor)

6.4 Transition Framing (Analyst → Analytics Engineer)

How to frame this role transition

You have 5+ years of advanced analytics experience (Power BI, DAX, SQL, commercial KPIs in FMCG). This project demonstrates you can go upstream — from consuming clean data to building the pipelines that create it. The SunCulture context shows you understand the business problems that drive data engineering decisions, not just the technical execution.

6.5 Suggested Talking Points for Interviews

- "I started by mapping all of SunCulture's data sources to understand what existed before designing anything. The IoT telemetry turned out to be the foundation for three other domains — credit, carbon, and agronomy — which shaped the entire warehouse design."
- "The carbon MRV pipeline was the most technically demanding piece. Every VCU sold to EDF Trading requires audit-grade evidence — so I designed the pipeline with complete lineage from raw device event to verified emission reduction, reproducible by a third-party auditor."
- "I used my existing Power BI expertise to prototype what the consuming layer would look like first, then worked backwards to define what the data model needed to deliver. That's a more pragmatic approach than building pipelines without knowing what the output looks like."
- "The credit data pipeline is high-stakes in a way that most analytics projects aren't. A delay or error in PAR reporting affects Bridgin covenant compliance and investor confidence — so I designed the M-Pesa payment event pipeline with near-zero latency and explicit SLA monitoring."

🔨 7. How to Recreate This Project

Step-by-step guide to rebuild the demo from scratch

Step 1 — Research phase

- Research SunCulture via their website, press releases, job postings, and Bridgin/Verra announcements
- Build the enterprise data inventory: map every department to its data entities, fields, and source systems (use the inventory in Section 2 as a template)
- Identify the data architecture: ingestion pattern, transformation logic, and serving layer for each domain

Step 2 — Data model design

- Design the star schema: identify fact tables (repayments, device_events, sales_orders, carbon_credits) and dimensions (farmer, product, date, geography)
- Define the shared farmer_id key that links all fact tables
- Document the grain of each fact table (e.g., fact_repayments = one row per payment event)
- Plan the 4-4-5 calendar dimension if the company uses it

Step 3 — Build the HTML prototypes

- Create one HTML file per report page. Use Chart.js (CDN) for all charts
- Write a genData() function that produces randomised but realistic data on each page load
- Include a Refresh button that re-calls genData() and rebuilds all charts
- Use consistent colour scheme: primary green #00A86B, amber #FFB800, coral #FF4757, dark #0A1628
- Build KPI cards with coloured top border (--accent CSS variable), value, sub-label, and delta badge

Step 4 — Build the master App shell

- Create index.html with a persistent sidebar navigation and topbar
- Use an <iframe> to embed each report page with a toolbar (back button, refresh, open in tab)
- Show a landing page with report cards when no report is selected
- Add data architecture notes on the landing page to reinforce the engineering narrative

Step 5 — Recreate in actual Power BI (for the real portfolio)

- Connect Power BI to sample data files (CSV or Excel) modelled to match the star schema design
- Build the same 6 pages with matching KPIs, charts, and layout as the HTML prototypes
- Configure Row-Level Security: county role for sales, portfolio slice for credit/investor
- Publish to Power BI Service and create an App workspace that mirrors the HTML app shell
- Document the data lineage from source → warehouse → Power BI in a separate architecture diagram

Step 6 — Write the project case study

- One-page summary: problem → approach → deliverables → skills demonstrated
- Include a simplified architecture diagram (can use draw.io or Lucidchart)
- Link to the live HTML demo (can host for free on GitHub Pages or Netlify)
- Reference the specific data engineering challenges: IoT streaming, audit-grade MRV, real-time PAR monitoring

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